

Chin-Pei He

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ORCID: [0009-0004-8913-9179](#) | Web of Science ResearcherID: PLC-4893-2026

RESEARCH PROFILE

Information Management researcher working toward responsible and auditable AI for human centred decision support, particularly in healthcare and organisational contexts. My research examines how explanations, risk assessment, human review, workflow design, and verification can support effective decision making while keeping AI enabled processes interpretable, reviewable, and accountable. My work spans process mining, simulation, clinical decision support, AI governance, health information systems, and computational verification, with current research experience at the Center for Neuropsychiatric Research, National Health Research Institutes.

EDUCATION

National Taipei University of Nursing and Health Sciences

M.S. in Information Management

Taiwan
Sep. 2024 to Jul. 2026

- Thesis: *Optimizing Carbon Emission Audit Processes Using Process Mining and Simulation*
- Research integrated process mining, discrete event simulation, explainable decision support, and human review to evaluate and redesign governance critical workflows.
- Relevant coursework: Machine Learning, Artificial Intelligence, Statistics, Data Mining, Operations Research, and Research Methods.

Chihlee University of Technology

B.S. in Business Technology

Taiwan
2020 to 2024

RESEARCH INTERESTS

Responsible and auditable AI, human centred decision support, human oversight, appropriate reliance, explainable AI, AI governance, clinical decision support, healthcare information systems, workflow analytics, process mining, simulation, and institutional accountability.

PUBLICATIONS

1. **AIDLC FMEA: Explainability Integrated Lifecycle Risk Assessment for AI Governance.**
Chin-Pei He, Yu-Ru Pan, and Kuo-Chung Chu.
IEEE Access, 2026. *Published online.*
DOI: [10.1109/ACCESS.2026.3701925](#).
2. **AIDLC Governance Indicator Framework: A Lifecycle Based Approach to Institutional AI Governance.**
Chin-Pei He, Yu-Ru Pan, and Kuo-Chung Chu.
AI & Society, 2026. *Published online.*
DOI: [10.1007/s00146-026-03173-9](#).

ACCEPTED CONFERENCE PAPERS

1. **Learning Promise–Evidence Alignment from Evidence Swaps without Misleading Labels: Team_10858 at the NTCIR-19 AI CUP-VeriPromise ESG Track.**
Tzu-Han Chuang, Chin-Pei He, Chih-Chung Ma, Hsin-Yao Lin, Yu-Hsuan Lui, and Chun Fu.
Accepted for the NTCIR-19 AI CUP-VeriPromise ESG Track, 2026.

MANUSCRIPTS UNDER REVIEW OR REVISION

1. **A Clinical Decision-Support Visualization System for Language Sample Analysis in Chinese-Speaking Children: Development and Usability Evaluation.**
Szu-Ming Huang, Chin-Pei He, Yu-Ru Pan, and Kuo-Chung Chu.
Manuscript under review at Computer Methods and Programs in Biomedicine; preprint available at SSRN, 2026.
SSRN DOI: [10.2139/ssrn.7331050](#).
2. **A Quasi 3D U Net Modeling Framework for Breast Tumor Segmentation in Dynamic Contrast Enhanced MRI.**
Han-Yu Chia, Chin-Pei He, Yu-Ru Pan, Tien-Yu Chang, and Kuo-Chung Chu.
Manuscript under revision for Journal of Medical and Biological Engineering.
3. **A Visualized Multi Label Disease Prediction Framework for Enhancing Health Decision Making and Reducing Information Asymmetry.**
Hoong-Seng Jao, Jui-Nan Cheng, Kuo-Chung Chu, Chin-Pei He, and Yu-Ru Pan.
Manuscript submitted to Health and Technology.
4. **Study on the Relationship Between Breast Cancer and Air Pollution Factors.**
Kuo-Chung Chu, Min-Yang Xiao, Chin-Pei He, and Yu-Ru Pan.
Manuscript under review or revision for Spatial and Spatio-temporal Epidemiology.

RESEARCH EXPERIENCE

Research Project Assistant

Center for Neuropsychiatric Research, National Health Research Institutes

Aug. 2026 to Present
Taiwan

- Support neuropsychiatric and public health research projects involving health data organisation, literature synthesis, research

documentation, and workflow based data preparation.

- Participate in health information standardisation and data mapping work, including the analysis of clinical and administrative data structures for interoperable research workflows.
- Assist in developing reproducible workflows for managing, transforming, and documenting health and biomedical research data.
- Collaborate with multidisciplinary researchers on project execution, data interpretation, methodological implementation, and preparation of research outputs.
- Develop practical experience in applying information management, data analytics, and decision support methods in real world healthcare and neuropsychiatric research environments.

Graduate Researcher

2024 to 2026

National Taipei University of Nursing and Health Sciences

Taiwan

- Conducted research under the supervision of Prof. Kuo-Chung Chu on decision support, workflow redesign, process mining, simulation, explainable systems, and technology governance.
- Developed lifecycle based governance and risk assessment frameworks, resulting in first author publications in *IEEE Access* and *AI & Society*.
- Investigated how explanations, risk assessment, and human review can be integrated into decision processes to support transparency, accountability, and reviewable decision making.
- Modelled baseline and redesigned workflows using BPMN and BIMP, generated event logs, and evaluated cycle time, waiting time, resource utilization, and makespan.
- Applied SHAP based explanations to support interpretable decision triage and human review.
- Combined workflow analytics and decision support methods to identify bottlenecks, human intervention points, and opportunities for process improvement.
- Supported manuscript preparation, journal submission, revision responses, literature synthesis, and research documentation.
- Mentored junior lab members in literature organization, research preparation, and academic writing.

TEACHING AND PROFESSIONAL EXPERIENCE

STEM Mechanical Building Instructor, Part Time

Jul. 2025 to Dec. 2025

Orange Apple Programming Education Center

Taiwan

- Delivered hands on educational activities using mechanical building systems and powered construction models.
- Guided students in structural design, mechanical principles, model construction, and problem solving.
- Provided individualized learning guidance and supported classroom preparation and educational activity implementation.

Programming Teaching Assistant

Fall 2023

Chihlee University of Technology

Taiwan

- Assisted with programming instruction, debugging, and student support.
- Helped students understand programming exercises and complete course assignments.

RESEARCH METHODS AND TECHNICAL SKILLS

- **Programming and Data Analysis:** Python, pandas, NumPy, scikit learn, exploratory data analysis, classification, regression, model evaluation, and quantitative analysis
- **Workflow Analytics and Process Mining:** BPMN, PM4Py, ProM, process discovery, event log analysis, bottleneck identification, workflow KPI evaluation, and process redesign
- **Simulation and Decision Support:** BIMP, discrete event simulation, scenario analysis, resource utilization analysis, decision support modelling, and workflow based evaluation
- **Interpretable and Reviewable Systems:** SHAP, global and local explanations, feature attribution, interpretable decision support, human review, and reviewable decision processes
- **Healthcare and Biomedical Research:** health data preparation, clinical decision support visualization, literature synthesis, research workflow development, biomedical research documentation, and health information standardisation
- **Research Communication:** manuscript preparation, reviewer response drafting, literature organization, journal submission, preprint preparation, and academic documentation

PEER REVIEW AND ACADEMIC SERVICE

- Verified peer reviewer with 10 peer review records on the Web of Science Researcher Profile.
- Served as a reviewer for *IEEE Access*, *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, and *Computer Methods in Biomechanics and Biomedical Engineering: Imaging & Visualization*.
- Reviewed interdisciplinary studies involving trustworthy AI, technology governance, clinical decision support, medical vision language models, reinforcement learning, computational modelling, biomedical imaging, intelligent systems, and methodological design.

RESEARCHER IDENTIFIERS

- **ORCID:** [0009-0004-8913-9179](https://orcid.org/0009-0004-8913-9179)
- **Web of Science ResearcherID:** PLC-4893-2026
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